```
!pip install chromadb
!pip install qdrant-client
!pip install pinecone
!pip install sentence-transformers
!pip install pymorphy3
!pip install pandas
!pip install numpy
!pip install tqdm
```

Показать скрытые выходные данные

```
import chromadb
from qdrant_client import QdrantClient
from qdrant_client.models import Distance, VectorParams, PointStruct

from pinecone import Pinecone, ServerlessSpec

import numpy as np
import pandas as pd
import re
from tqdm import tqdm
import time

from sentence_transformers import SentenceTransformer
import pymorphy3

import json
import matplotlib.pyplot as plt
from google.colab import userdata
```

```
model = SentenceTransformer("all-MiniLM-L6-v2")
model_size = 384
```

Показать скрытые выходные данные

```
morph = pymorphy3.MorphAnalyzer()
```

```
with open('task_data.json', 'r', encoding='utf-8') as f:
    dataset = json.load(f)

facts = dataset['facts']
queries = dataset['queries']

texts = [f["text"] for f in facts]
topics = [f["topic"] for f in facts]

print(f"Загружено фактов: {len(facts)}")
```

```
Загружено фактов: 100
```

Функция базовой предобработки текста

```
def preprocess_raw(text):
    return text
```

Функция NLP-предобработки: токенизация + лемматизация

```
def preprocess_nlp(text):
    words = re.findall(r'\w+', text.lower())
    lemmas = [morph.parse(w)[0].normal_form for w in words]

    return " ".join(lemmas)
```

Функция добавления тематических тегов

```
def preprocess_tags(text, topic):
    tag_map = {
        "temperature":"#жара #температура",
        "humidity":"#влажность #конденсат",
        "raw_material":"#сырье #металл",
        "vibration":"#вибрация #станок",
        "coolant":"#сож #охлаждение",
        "robots":"#роботы #agv",
        "inventory":"#склад #сканер",
```

```
            "refrigeration":"#холод #камера",
            "packaging":"#упаковка #коробка",
            "delivery":"#доставка #логистика"
        }

        tags = tag_map.get(topic, "")

        return text + " " + tags
```

Функция подготовки текста фактов с использованием выбранного метода предобработки

```
    def prepare_texts(method):
        processed = []
        for f in facts:
            text = f["text"]
            topic = f["topic"]

            if method == "raw":
                processed.append(preprocess_raw(text))
            elif method == "nlp":
                processed.append(preprocess_nlp(text))
            elif method == "tags":
                processed.append(preprocess_tags(text, topic))

        return processed
```

Функция подготовки текста запроса

```
    def prepare_query(method, query):
        if method == "raw":
            return preprocess_raw(query)
        elif method == "nlp":
            return preprocess_nlp(query)
        elif method == "tags":
            return query
```

Функция вычисления embedding для списка текстов

```
    def embed(texts):
        return model.encode(texts, normalize_embeddings=True)
```

Функция оценки точности поиска (= количество релевантных фактов / количество результатов)

```
    def evaluate(returned_ids, expected_topic):
        correct = 0
        for id in returned_ids:
            if topics[id] == expected_topic:
                correct += 1
        return correct / len(returned_ids)
```

Функция тестирования ChromaDB: загрузка данных, выполнение запросов, измерение точности и времени

```
    def test_chromadb(texts, embeddings, query_embeddings, method):
        client = chromadb.Client()

        try:
            client.delete_collection(name="facts_chroma")
        except:
            pass

        collection = client.create_collection("facts_chroma")
        ids = [str(i) for i in range(len(texts))]

        collection.add(
            documents=texts,
            embeddings=embeddings.tolist(),
            ids=ids
        )

        scores = []
        times = []
        for q, q_emb in zip(queries, query_embeddings):
            start = time.time()

            results = collection.query(
```

```
            query_embeddings=[q_emb.tolist()],
            n_results=10
        )

        elapsed = time.time() - start
        times.append(elapsed)

        returned_ids = [int(i) for i in results["ids"][0]]
        score = evaluate(returned_ids, q["expected_topic"])
        scores.append(score)

    return np.mean(scores), np.mean(times)
```

Функция тестирования Qdrant: создание коллекции, загрузка векторов, поиск и измерение точности и времени

```
def test_qdrant(texts, embeddings, query_embeddings, method, model_size):
    client = QdrantClient(":memory:")
    collection_name = "facts_qdrant"

    client.recreate_collection(
        collection_name=collection_name,
        vectors_config=VectorParams(size=model_size, distance=Distance.COSINE)
    )

    points = [
        PointStruct(
            id=i,
            vector=emb.tolist(),
            payload={"text": texts[i]}
        ) for i, emb in enumerate(embeddings)
    ]

    client.upsert(collection_name=collection_name, points=points)

    scores = []
    times = []
    for q, q_emb in zip(queries, query_embeddings):
        start = time.time()

        hits = client.query_points(
            collection_name=collection_name,
            query=q_emb,
            limit=10
        ).points

        elapsed = time.time() - start
        times.append(elapsed)

        returned_ids = [hit.id for hit in hits]
        score = evaluate(returned_ids, q["expected_topic"])
        scores.append(score)

    return np.mean(scores), np.mean(times)
```

Функция тестирования Pinecone: создание индекса, загрузка данных, выполнение поиска и измерение точности и времени

```
def test_pinecone(texts, embeddings, query_embeddings, method, model_size):
    pc = Pinecone(api_key=userdata.get('pinecone_api_key'))
    index_name = "facts-pinecone"

    if index_name in pc.list_indexes().names():
        pc.delete_index(index_name)

    pc.create_index(
        name=index_name,
        dimension=model_size,
        metric="cosine",
        spec=ServerlessSpec(
            cloud="aws",
            region="us-east-1"
        )
    )

    index = pc.Index(index_name)

    vectors = [
        (str(i), embeddings[i].tolist(), {"text": texts[i]})
        for i in range(len(texts))
    ]
```

```
    index.upsert(vectors=vectors)

    scores = []
    times = []

    for q, q_emb in zip(queries, query_embeddings):
        start = time.time()

        results = index.query(
            vector=q_emb.tolist(),
            top_k=10
        )

        elapsed = time.time() - start
        times.append(elapsed)

        returned_ids = [int(m["id"]) for m in results["matches"]]
        score = evaluate(returned_ids, q["expected_topic"])
        scores.append(score)

    return np.mean(scores), np.mean(times)
```

Функция тестирования 3х исследуемых баз данных, сбор результатов измерений

```
def test_all(model_size):
  preprocess_methods = ["raw", "nlp", "tags"]
  results = []

  for m in preprocess_methods:
      print("Testing:", m)
      texts_processed = prepare_texts(m)
      queries_processed = [prepare_query(m, q["query"]) for q in queries]

      fact_embeddings = embed(texts_processed)
      query_embeddings = embed(queries_processed)

      chroma_score, chroma_time = test_chromadb(texts_processed, fact_embeddings, query_embeddings, m)
      qdrant_score, qdrant_time = test_qdrant(texts_processed, fact_embeddings, query_embeddings, m, model_size)
      pinecone_score, pinecone_time = test_pinecone(texts_processed, fact_embeddings, query_embeddings, m, model_size)

      results.append({
          "method": m,
          "chromadb_accuracy": chroma_score,
          "chromadb_time": chroma_time,
          "qdrant_accuracy": qdrant_score,
          "qdrant_time": qdrant_time,
          "pinecone_accuracy": pinecone_score,
          "pinecone_time": pinecone_time
      })

  return results
```

Проведение эксперимента с первой embedding моделью "all-MiniLM-L6-v2"

```
results = test_all(model_size)
df = pd.DataFrame(results)
df
```

```
Testing: raw
/tmp/ipykernel_1373/1405104879.py:6: DeprecationWarning: `recreate_collection` method is deprecated and will be removed in t
  client.recreate_collection(
Testing: nlp
Testing: tags
```

	method	chromadb_accuracy	chromadb_time	qdrant_accuracy	qdrant_time	pinecone_accuracy	pinecone_time
0	raw	0.16	0.001859	0.16	0.000688	0.16	0.058133
1	nlp	0.19	0.001279	0.19	0.000598	0.19	0.054141
2	tags	0.30	0.001291	0.30	0.000518	0.30	0.056355

Далее: Создать код с переменной df New interactive sheet

График точности поиска в исследуемых базах данных при различных алгоритмах предобработки для первого эксперимента

```
df.set_index("method")[["chromadb_accuracy","qdrant_accuracy","pinecone_accuracy"]].plot(kind="bar")

plt.title("Vector DB accuracy comparison")
```

```
plt.ylabel("Accuracy")
plt.show()
```

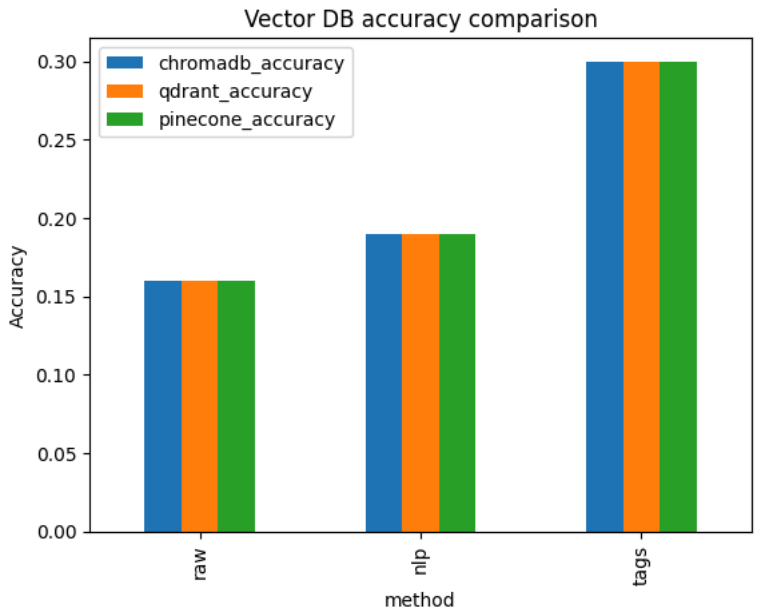
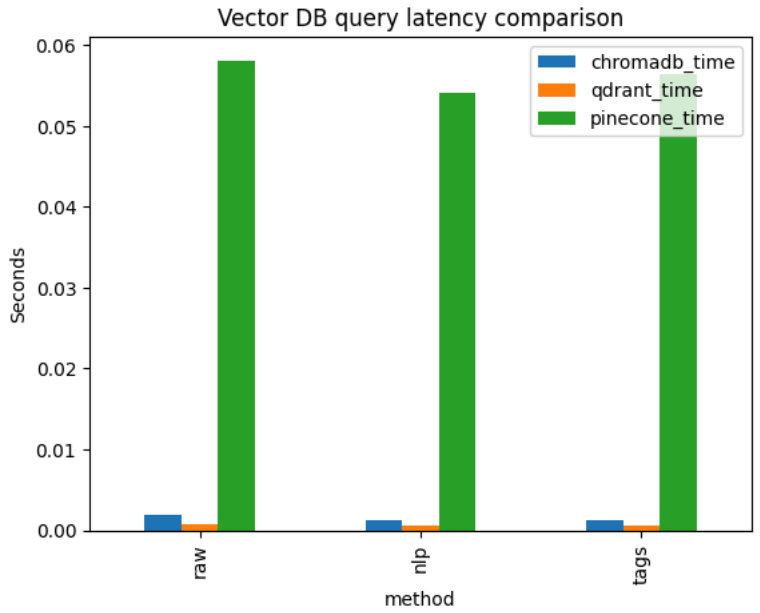


График времени обработки запроса в исследуемых базах данных при разных алгоритмах предобработки для первого эксперимента

```
df.set_index("method")[["chromadb_time","qdrant_time","pinecone_time"]].plot(kind="bar")

plt.title("Vector DB query latency comparison")
plt.ylabel("Seconds")
plt.show()
```



Проведение эксперимента со второй embedding моделью "intfloat/multilingual-e5-base"

```
model = SentenceTransformer("intfloat/multilingual-e5-base")
model_size = 768
results = test_all(model_size)

df = pd.DataFrame(results)
df
```

```
Loading weights: 100%                                    199/199 [00:00<00:00, 640.84it/s, Materializing param=pooler.dense.weight]
XLMRobertaModel LOAD REPORT from: intfloat/multilingual-e5-base
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Testing: raw
/tmp/ipykernel_1373/1405104879.py:6: DeprecationWarning: `recreate_collection` method is deprecated and will be removed in t
  client.recreate_collection(
Testing: nlp
Testing: tags
```

	method	chromadb_accuracy	chromadb_time	qdrant_accuracy	qdrant_time	pinecone_accuracy	pinecone_time
0	raw	0.44	0.001512	0.44	0.000831	0.44	0.048709
1	nlp	0.43	0.001403	0.43	0.000951	0.44	0.052210
2	tags	0.66	0.001424	0.66	0.001009	0.65	0.055412

Далее: Создать код с переменной df New interactive sheet

График точности поиска в исследуемых базах данных при различных алгоритмах предобработки для второго эксперимента

```
df.set_index("method")[["chromadb_accuracy","qdrant_accuracy","pinecone_accuracy"]].plot(kind="bar")

plt.title("Vector DB accuracy comparison. 1st ")
plt.ylabel("Accuracy")
plt.show()
```

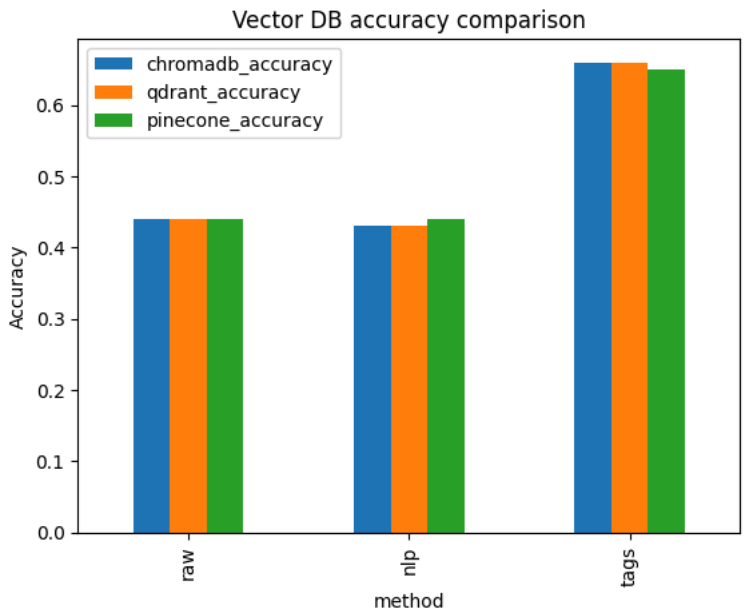
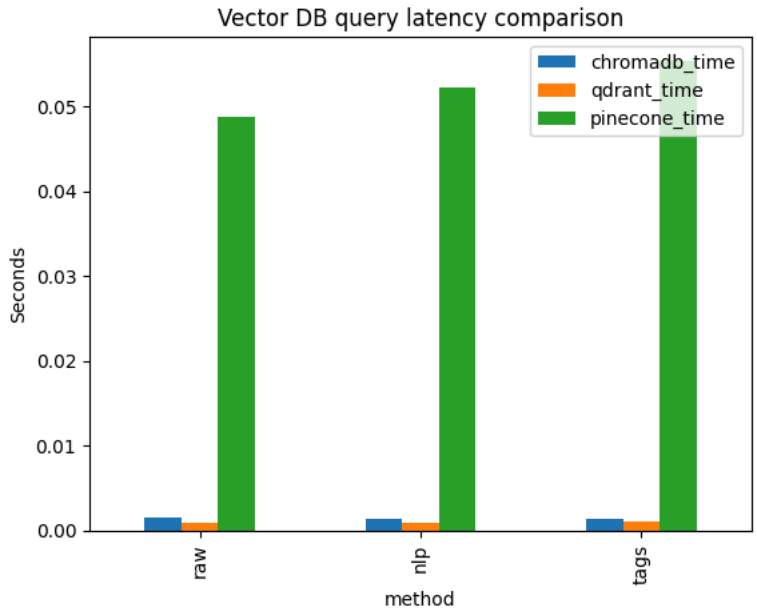


График времени обработки запроса в исследуемых базах данных при разных алгоритмах предобработки для второго эксперимента

```
df.set_index("method")[["chromadb_time","qdrant_time","pinecone_time"]].plot(kind="bar")

plt.title("Vector DB query latency comparison")
plt.ylabel("Seconds")
plt.show()
```


Vector DB query latency comparison
chromadb_time
qdrant_time
pinecone_time
Seconds
0.00
0.01
0.02
0.03
0.04
0.05
raw
nlp
tags
method